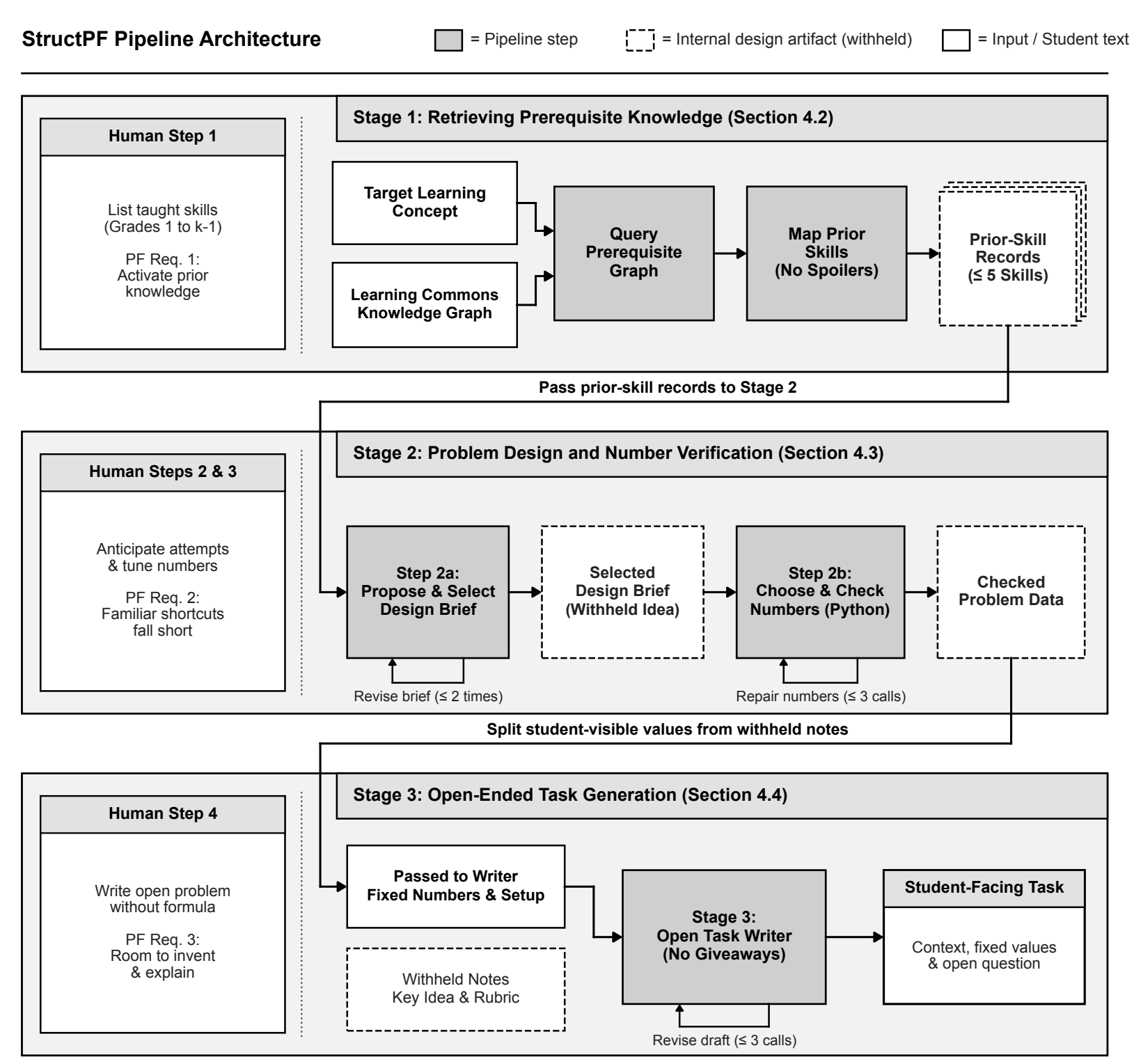
StructPF Pipeline Architecture
= Pipeline step
= Internal design artifact (withheld)
= Input / Student text
Human Step 1
List taught skills (Grades 1 to k-1)
PF Req. 1: Activate prior knowledge
Stage 1: Retrieving Prerequisite Knowledge (Section 4.2)
Target Learning Concept
Learning Commons Knowledge Graph
Query Prerequisite Graph
Map Prior Skills (No Spoilers)
Prior-Skill Records (≤ 5 Skills)
Pass prior-skill records to Stage 2
Human Steps 2 & 3
Anticipate attempts & tune numbers
PF Req. 2: Familiar shortcuts fall short
Stage 2: Problem Design and Number Verification (Section 4.3)
Step 2a: Propose & Select Design Brief
Selected Design Brief (Withheld Idea)
Step 2b: Choose & Check Numbers (Python)
Checked Problem Data
Revise brief (≤ 2 times)
Repair numbers (≤ 3 calls)
Split student-visible values from withheld notes
Human Step 4
Write open problem without formula
PF Req. 3: Room to invent & explain
Stage 3: Open-Ended Task Generation (Section 4.4)
Passed to Writer Fixed Numbers & Setup
Withheld Notes Key Idea & Rubric
Stage 3: Open Task Writer (No Giveaways)
Student-Facing Task
Context, fixed values & open question
Revise draft (≤ 3 calls)